Chemistry - Graduate Level Assessment 18 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The symbol for antimony is
   (a) Am
   (b) As
   (c) Ab
   (d) Au
   (e) Fe

2. A saturated solution of CaF_2 contains .00168 g of CaF_2 per 100 g of water. Determine theK_sp.
   (a) $2.11 \times 10^{-}8$
   (a) $2.73 \times 10^{-}8$
   (a) $1.08 \times 10^{-}8$
   (a) $3.92 \times 10^{-}8$
   (a) $4.77 \times 10^{-}8$

3. Write theorbitals, in the order of increasing energy, for hydrogen - like atoms.
   (a) 1 s < 2 p < 2 s < 3 p < 3 s < 3 d
   (b) 2 s < 2 p < 1 s < 3 s < 3 p < 3 d
   (c) 1 s < 2 s = 2 p < 3 s = 3 p = 3 d < 4 s = 4 p = 4 d = 4 f < 5 s = 5 p = 5 d = 5 f = 5 g
   (d) 2 s = 2 p < 1 s = 3 p = 3 d
   (e) 1 s < 3 s < 2 s < 5 s < 4 s

4. Use the van der Waals parameters for chlorine to calculate approximate values of the Boyle temperature of chlorine.
   (a) 1.25×10^3 K
   (b) 1.20×10^3 K
   (c) 1.41×10^3 K
   (d) 1.10×10^3 K
   (e) 1.65×10^3 K

5. The +1 oxidation state is more stable than the +3 oxidation state for which group 13 element?
   (a) Al
   (b) Tl
   (c) C
   (d) B
   (e) Si

True / False (10 marks)

Answer True or False

6. True or False: The equation x^2 - 5 x + 6 = 0 has solutions of x = 4 and x = 5 when solved using the quadratic formula.
7. True or False: Methylamine (CH 3 NH 2) can form hydrogen bonds due to the presence of a nitrogen atom with a lone pair of electrons.
8. True or False: The 13 C spectrum of 3-ethylpentane exhibits five distinct chemical shifts due to its unique molecular structure.
9. True or False: Covalent network bonding typically results in high melting points and poor electrical conductivity in substances.
10. True or False: The logarithm of 3^2 is equal to 3, representing the exponent directly without any calculations.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that influence the chemical shift in NMR spectroscopy and analyze why CH 3 Cl exhibits a 1 H resonance approximately 1.55 kHz from TMS in a 12.0 T magnetic field.
12. Discuss the process by which cobalt-60 is produced from cobalt-59, emphasizing the role of neutron bombardment in radiation therapy applications for cancer treatment.

End of Paper